KAMALESAN C

AI ENGINEER • GENERATIVE AI • AI AGENTS • PYTHON AUTOMATION

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PROFESSIONAL SUMMARY

AI-focused engineer with **7.4 years of engineering experience** across Generative AI application development, Python automation, automotive embedded systems, and software engineering. Hands-on experience architecting autonomous AI agents, LLM pipelines, and backend integrations using **Gemini and OpenAI APIs, Python, FastAPI, REST APIs, PostgreSQL, SQLAlchemy, Redis, and Docker**. Proven expertise implementing human-in-the-loop workflows, MCP concepts, deterministic validation gates, and automated test orchestration. Open to worldwide remote roles and international relocation.

CORE TECHNICAL SKILLS

Generative AI & LLMs:	Gemini API, OpenAI API, LLM Application Architecture, Prompt Engineering, Context Engineering, RAG Concepts, Function Calling / Tool Calling
AI Agents & Workflow:	Autonomous Agents, Multi-Agent Orchestration, Model Context Protocol (MCP), n8n Automation, Human-in-the-Loop Validation, Audit Trails
Backend & APIs:	Python, FastAPI, RESTful APIs, PostgreSQL, SQLAlchemy, Redis, SQLite, Microservices, JWT Authentication
Testing & Automation:	Python Test Automation, Playwright, TraceTronic ECU-TEST, HIL & SIL Testing, Diagnostic Protocols (CAN, UDS, OBD-II, OBDonUDS)
DevOps & Cloud:	Docker Containerization, Git & GitHub, Google Cloud (Cloud Run), Linux, Asynchronous Task Pipelines
Frontend & UI:	React, API-Driven Operational Dashboards, Risk Visibility Interfaces, Interactive Web UIs

KEY AI & AUTOMATION PROJECTS

EpiAgent PMO — AI Project Governance & Risk Platform

Google Cloud Run | GitHub

Python, FastAPI, GitLab API, Gemini API, Risk Engine, Interactive Dashboard

- Designed an enterprise AI platform that ingests project and repository data via GitLab API to generate portfolio governance and risk visibility.
- Integrated Gemini API to produce executive summary reports, risk interpretations, blocked-task detection, and action recommendations.
- Implemented automated scoring algorithms for project health, completion rates, and critical issue alerts with production REST endpoints.

AI-Powered SIL Test Automation Platform

Python | MCP | ECU-TEST

Gemini API, Antigravity, MCP Tool Interfaces, Python, FastAPI, TraceTronic ECU-TEST, RSC/SIL

- Designed an AI orchestration layer interfacing with automotive software-in-the-loop (SIL) test execution pipelines to cut manual setup effort.
- Built structured agent workflow: Context Retrieval → Requirement Parsing → Reusable Test Discovery → Draft Generation → Engineer Approval → Controlled Execution → Failure Analysis.
- Enforced strict human-in-the-loop approval gates for WRITE/EXECUTE actions, maintaining deterministic audit logs and evidence-first reports.

Automation Job Search & Applying Agent

Python | Playwright | LLM APIs

Python, Playwright, FastAPI, PostgreSQL, Gemini/OpenAI APIs, Excel Tracker

- Built an autonomous application assistant that scrapes job portals (LinkedIn, EU Remote Jobs) and categorizes opportunities by role taxonomy.
- Implemented dynamic cover letter and profile customization using LLMs, plus Playwright-powered automated ATS form filling (Greenhouse, Lever, generic ATS) with interactive human review modes.

AI Instagram Publisher Agent

FastAPI | React | Docker

Python, FastAPI, React, PostgreSQL, Redis, Docker, Meta Graph API, OpenAI/Gemini

- Developed multi-account social media publishing engine featuring scheduled queues, media management, and automated caption/hashtag generation.

PROFESSIONAL EXPERIENCE

Test Engineer (HIL & SIL) • Capgemini Technology Services — Oct 2021 – Present | Coimbatore, India

Client: Bosch Global Software Technologies • Mercedes-Benz Powertrain Management System Testing

- Validate powertrain management ECUs for Mercedes-Benz diesel and gasoline engine configurations across multiple variants.
- Develop and execute Python-based automated HIL/SIL test scripts using TraceTronic ECU-TEST, improving execution efficiency by ~30%.
- Perform diagnostic validation using CAN, UDS, OBD-II, and OBDonUDS protocols; simulate networks with CANoe, CANalyzer, INCA, and UDE.
- Spearhead structured failure analysis across signals, ECU communications, and test configurations; apply LLM-assisted workflows to accelerate requirement interpretation and root-cause analysis.

Embedded Software Engineer • GRID RGD — May 2019 – Oct 2021 | Bangalore, India

Embedded C, PIC, 8051 Microcontrollers, GSM, Zigbee Wireless, Power Management

- Developed Embedded C logic for GSM/SMS-based remote appliance automation, relay control, and status monitoring.
- Engineered 8051-based monitoring and Zigbee wireless communications for renewable energy switching and power management.

CERTIFICATIONS

- **Generative AI Mastermind** — OutSkill
- **AI Foundation** — Udemy
- **n8n Masterclass: AI Automations & Workflows** — Udemy
- **Prompt Engineering Frameworks & Methodologies** — Udemy

EDUCATION & DETAILS

- **Master of Computer Applications (M.C.A)**
  Bharathidasan University | 2024 – 2026
- **Languages:** English (Professional), Tamil (Native)
- **Mobility:** Open to Worldwide Remote & Visa Sponsorship